Senior Mathematics Enrichment

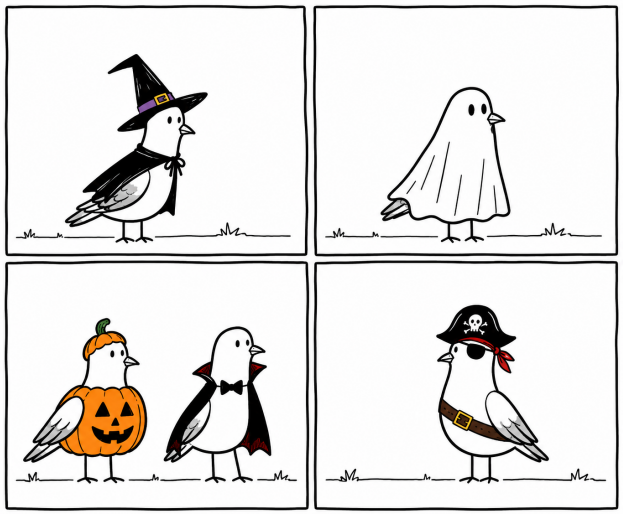
Combinatorics

2 The Pigeonhole Principle, Permutations and Combinations

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Today...

1. Warm-up question
2. Counting Pigeons
   - Trick? Or treat?
   - Pigeonholes
   - Target practice
   - The Pigeonhole Principle
   - Tricky!
   - What's the difference
   - One last one...
3. Permutations
   - Best dressed!
   - Factorials
4. Combinations
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   - Binomials
   - Pascal's identity
   - Another one
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Warm-up question

You are given **12** identical-looking metal balls and an old-fashioned balance weighing scale with two baskets. If you place objects in both baskets, the scale can tell you which basket (if either) has heavier contents. One of the balls has a different weight from the others. Using the balance, you must find the odd-one-out, and determine if it is lighter or heavier.
How can this be done with as few weighing as possible?

Trick or Treat!

You've been out trick-or-treating. Your bag contains some bars of chocolate. You reach into the bag to get some, but you can't look into the bag (it is dark!)...

(a) The bag contains (just) Twix and Kit-kats. How many bars must be taken from the bag in order to be certain of having two of the same type?

(b) If the bag also has Mars Bars, how many bars must be drawn in order to be certain of having two of the same type?

(c) How many bars must be taken from the bag in Part (a) in order to be certain of having *two* pairs of matching bars?

(d) How many bars be taken from the bag of Part (a) in order to be certain of having *four* of the same type?

A group of people are how trick-or-treating... How many must be in the group in order to be certain that

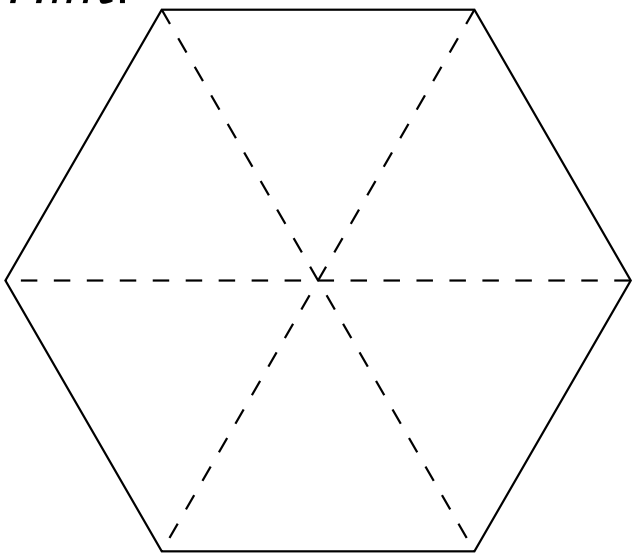
1. Two have their birthdays in the same month?
2. Three have their birthdays in the same month?
3. Two have their birthdays on the same day?

Finally, some pigeons!

10 boxes contain a total of 50 pigeons. Every pair of boxes contains a total of at least 8 pigeons. What is the least number of pigeons that must be in any box?
Hint: Is it possible for a box to have no pigeons?

A target has the shape of a regular hexagon with side length 1m. Hitting the target makes a mark. If the target is hit seven times, show that there are two marks at most 1 metre apart.

Hint:



So far, we've been using the following idea:

The Pigeonhole Principle

If n items are put into m containers, with $n > m$, then at least one container must contain more than one item.



A slightly trickier problem

101 distinct points are chosen in a 10 metre by 10 metre square. Prove that some three of these points form a triangle whose area is no more than $1m^2$.

1. Take any three whole numbers. Show that among the three are two whose difference is an even number.
2. Take any six whole numbers. Show that among the six numbers are two whose difference is exactly divisible by 5.
3. Take any twelve two-digit whole numbers. Show that among the twelve of them there must be two whose difference is a number of the form aa, i.e. the first and second digits in this difference are the same.

An old IMO question

In a tennis tournament with 50 players, each player plays a match against each of the others. Every player wins at least one match. Prove that there must be three players A, B and C, so that A beat B, B beat C and C beat A.

4 people enter a competition for most scary Halloween consume.
There will be prizes for 1st, 2nd and 3rd.
How many ways can the prizes be awarded?



Factorial

If you have n objects then the number of different ways of ranking them is

$$n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1 = n! \quad (\text{“}n \text{ factorial}).$$

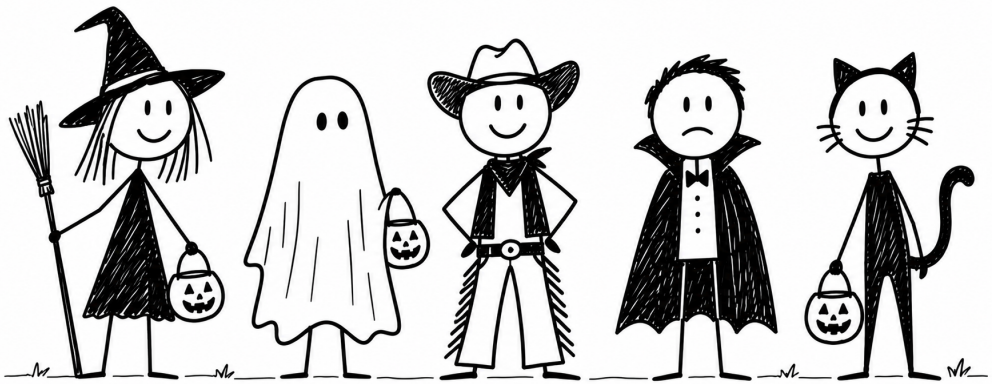
As n increases, $n!$ increases very quickly. Did you know that the age of the universe is less than 20! seconds old?.

Permutations

In general, if you have n objects, the number of ways of ranking $k \leqslant n$ of them is

$$\frac{n!}{(n-k)!}.$$

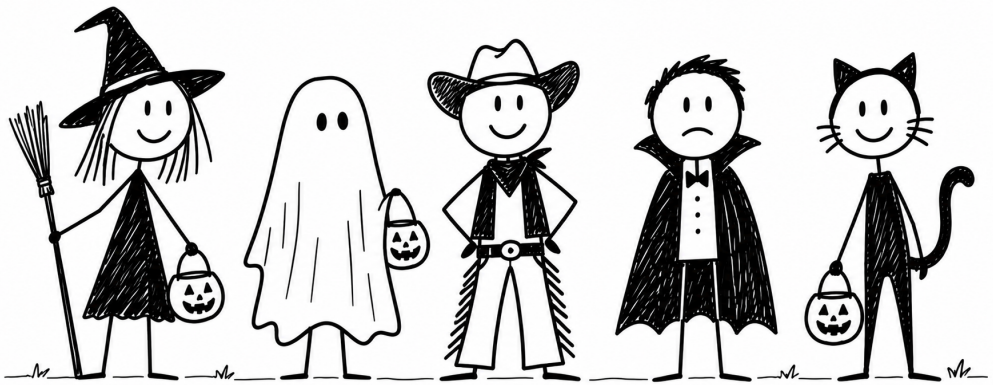
We call this a **permutation**.



Combinations

In a **permutation**, order matters (like giving put a 1st, 2nd and 3rd place prize). With a **combination** order does *not* matter (like giving the same prize to the top 3).

5 kids are going T&T'ing; there are only 3 baskets. How many ways can the baskets be allocated, if each gets as most one?



Combinations

We use $\binom{n}{k}$ to denote the number of ways of choosing k items from n.

A fact that we will ignore, for now, about about the Binomial coefficient formula

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}. \tag{1}$$

Let's make some lists!

Binomial coefficient

$\binom{n}{k}$ is also called the "binomial coefficient" because the coefficient of $a^k b^{n-k}$ in $(a+b)^n$ is $\binom{n}{k}$. That is

$$(a+b)^n = a^n + \binom{n}{n-1} a^{n-1} b + \binom{n}{n-2} a^{n-2} b^2 + \cdots + \binom{n}{1} a^1 b^{n-1} + b^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}. \quad (2)$$

Pascal's Identity

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Another Identity

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

Finally, lets prove that...

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}. \tag{3}$$

Here are some other identities. *Can you prove the following?*

1. $\binom{n}{k} = \binom{n}{n-k}$.
2. $\binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n}$.
3. $1 + 2 + 3 + \cdots + n = \binom{n+1}{2}$.

4. $\binom{x+y}{2} - \binom{x}{2} - \binom{y}{2} = xy$.

5. $\binom{n}{2}\binom{n-2}{k-2} = \binom{n}{k}\binom{k}{2}$.

6. $$\binom{2}{2}\binom{n}{2} + \binom{3}{2}\binom{n-1}{2} + \binom{4}{2}\binom{n-2}{2} + \cdots + \binom{n}{2}\binom{2}{2}1 = \binom{n+3}{5}$$

7. $$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0.$$

8. Find an simple expression for $\displaystyle\sum_{k=0}^{n} \binom{n}{k}$.

9. $\binom{m+n}{k} = \sum_{r=0}^{k} \binom{m}{k-r}\binom{n}{r}$.

10. $\binom{n-1}{k} - \binom{n-1}{k-1} = \frac{n-2k}{n}\binom{n}{k}$.

11. How many ways can you write n as the sum of r non-negative integers, where order matters? E.g, three of the ways of writing $n = 5$ as $r = 3$ integers are
$5 = 0 + 1 + 4$, $5 = 1 + 0 + 4$, $5 = 1 + 2 + 2$.